ADMINISTRATION SYSTÈME

Elasticsearch & Kibana

Suite Elastic — Installation et Configuration sous Debian 12

Auteur	**bthiery (Kalandy)**
Système	**Debian 12**
Version	**Elastic 9.x**
Service	**Elasticsearch + Kibana**
Accès	**https://192.168.193.28:5601**

Partie I - Objectif et Prérequis

Cette documentation décrit l'installation, la configuration et la mise en service de la suite Elastic (Elasticsearch et Kibana) version 9.x sur un serveur Debian 12. L'objectif est d'obtenir une plateforme centralisée de logs et de supervision fonctionnelle et sécurisée (HTTPS).

Prérequis

- Une machine Debian 12 à jour
- Les droits administrateur (root)
- Un accès Internet pour récupérer les paquets

Partie II - Optimisation du Système (Kernel)

Elasticsearch nécessite une configuration spécifique de la mémoire virtuelle (vm.max_map_count) pour fonctionner correctement.

2.1 - Modifier la configuration mémoire

➢ **Action : Modifier le paramètre kernel en live, puis le rendre persistant.**

```
sysctl -w vm.max_map_count=262144
echo 'vm.max_map_count=262144' | tee -a /etc/sysctl.conf
sysctl -p
```

✓ **Résultat :** *Résultat attendu : vm.max_map_count = 262144*

Partie III - Préparation de l'Environnement

3.1 - Mise à jour et installation des outils

```
apt update && apt -y full upgrade
apt install -y apt-transport-https gnupg wget curl systemd-timesyncd
```

3.2 - Configuration du fuseau horaire et NTP

```
timedatectl set-ntp true
timedatectl set-timezone "Europe/Paris"
timedatectl
```

3.3 - Activer le service de temps

```
systemctl enable --now systemd-timesyncd.service
systemctl start systemd-timesyncd.service
systemctl status systemd-timesyncd.service
```

Partie IV - Installation du Dépôt Officiel Elastic

4.1 - Récupérer la clé GPG

```
wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --
dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg
ls -l /usr/share/keyrings/elasticsearch-keyring.gpg
```

4.2 - Ajouter le dépôt APT

```
VERS=9.x
echo "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg]
https://artifacts.elastic.co/packages/${VERS}/apt stable main" | tee
/etc/apt/sources.list.d/elastic-${VERS}.list
cat /etc/apt/sources.list.d/elastic-${VERS}.list
```

Partie V - Installation et Configuration d'Elasticsearch

5.1 - Installer le paquet

⚠ **Attention :** *À la fin de l'installation, noter le mot de passe du superutilisateur elastic affiché dans le bloc de sécurité.*

```
apt update && apt install -y elasticsearch
export ELASTIC_PASSWORD="*TFAS8Ht6wKd1wort=bD"
```

5.2 - Modifier la configuration réseau

➤ **Action : Éditer le fichier de configuration : nano /etc/elasticsearch/elasticsearch.yml**

```
# Section Network :
network.host: 0.0.0.0
```

5.3 - Augmenter le délai de démarrage

```
mkdir -p /etc/systemd/system/elasticsearch.service.d
echo -e "[Service]\nTimeoutStart=180" | tee
/etc/systemd/system/elasticsearch.service.d/startup-timeout.conf
systemctl daemon-reload
systemctl show elasticsearch | grep ^Timeout
```

5.4 - Démarrer le service

```
systemctl enable elasticsearch.service
systemctl start elasticsearch.service
systemctl status elasticsearch.service
lsof -i -P -n | grep LISTEN | grep 9200
```

5.5 - Test de connexion

```
curl -k -u "elastic:${ELASTIC_PASSWORD}" https://localhost:9200
```

✓ **Résultat :** *Réponse JSON confirmant la connexion à Elasticsearch en HTTPS.*

Partie VI - Installation et Configuration de Kibana

6.1 - Installer Kibana

```
apt update && apt install -y kibana
```

6.2 - Configuration initiale

➢ **Action : Éditer le fichier : nano /etc/kibana/kibana.yml**

```
1 #migrations.batchSize: 1000
2
3 # The maximum payload size for indexing batches of upgraded saved objects.
4 # To avoid migrations failing due to a 413 Request Entity Too Large response from Elasticsearch.
5 # This value should be lower than or equal to your Elasticsearch cluster's `http.max_content_length`
6 # configuration option. Default: 100mb
7 #migrations.maxBatchSizeBytes: 100mb
8
9 # The number of times to retry temporary migration failures. Increase the setting
10 # if migrations fail frequently with a message such as `Unable to complete the [...] step after
11 # 15 attempts, terminating`. Defaults to 15
12 #migrations.retryAttempts: 15
13
14 # ================== Search Autocomplete ==================
15 # Time in milliseconds to wait for autocomplete suggestions from Elasticsearch.
16 # This value must be a whole number greater than zero. Defaults to 1000ms
17 #unifiedSearch.autocomplete.valueSuggestions.timeout: 1000
18
19 # Maximum number of documents loaded by each shard to generate autocomplete suggestions.
20 # This value must be a whole number greater than zero. Defaults to 100_000
21 #unifiedSearch.autocomplete.valueSuggestions.terminateAfter: 100000
22
23 ### >>>>>>> BACKUP END: Kibana interactive setup (2026-01-06T15:29:50.292Z)
24
25 # This section was automatically generated during setup.
26 server.host: 0.0.0.0
27 elasticsearch.hosts: [https://192.168.193.28:9200]
28 logging.appenders.file.type: file
29 logging.appenders.file.fileName: /var/log/kibana/kibana.log
30 logging.appenders.file.layout.type: json
31 logging.root.appenders: [default, file]
32 pid.file: /run/kibana/kibana.pid
33 elasticsearch.serviceAccountToken: AAEAAWVzYXN0aWMva21lYW5hL2Vucm9sbC1wcm9jZXNzLXRva2VuLTE3Nj c3MTMzODk1OTQ6WnE4UWNRUmpUc1dXV1ltQ0tKME1IQQ
34 elasticsearch.ssl.certificateAuthorities: [/var/lib/kibana/ca_1767713390288.crt]
35 xpack.fleet.outputs: [{id: fleet-default-output, name: default, is_default: true, is_default_monitoring: true, type: elasticsearch, hosts: [https://192.168.193.28:9200], ca_trusted_fingerprint: 7fba03f7d9328cd04d762b9f96b86cacd47a2b0eed
36 xpack.encryptedSavedObjects.encryptionKey: 7dec6385e33db17bfd2f0b2e70f48270
37 xpack.reporting.encryptionKey: 817964eb013bfb7dd6ce3f9d4187ea01
38 xpack.security.encryptionKey: 65785b7084eedb8afa0c1606804bd9c1
```

```
# Ligne 11 :
server.host: "0.0.0.0"
# Ligne 43 :
elasticsearch.hosts: ["https://localhost:9200"]
```

6.3 - Démarrage du service

```
systemctl enable kibana
systemctl start kibana
journalctl -u kibana -f
```

```
42 server.publicBaseUrl: "https://192.168.193.28:5601/"
43 server.ssl.enabled: true
44 server.ssl.certificate: /etc/kibana/server.crt
45 server.ssl.key: /etc/kibana/server.key
```

Partie VII - Enrôlement et Sécurité de Kibana

7.1 - Générer le token d'enrôlement

```
/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s
kibana
```

7.2 - Récupérer le code de vérification

```
/usr/share/kibana/bin/kibana-verification-code
/usr/share/kibana/bin/kibana-encryption-keys generate
```

Copier les 3 lignes générées (xpack...) et les ajouter à la fin de /etc/kibana/kibana.yml.

```
nano /etc/kibana/kibana.yml
# Ajouter :
xpack.encryptedSavedObjects.encryptionKey:
7dec6385e33db17bfd2f0b2e70f48270
xpack.reporting.encryptionKey: 817964eb013bfb7dd6ce3f9d4187ea01
xpack.security.encryptionKey: 65785b7084eedb8afa0c1606804bd9c1
```

Partie VIII - Configuration SSL pour Kibana

8.1 - Récupérer le mot de passe du Keystore

```
/usr/share/elasticsearch/bin/elasticsearch-keystore show
xpack.security.http.ssl.keystore.secure_password
```

8.2 - Extraire les certificats PKCS12

```
openssl pkcs12 -in /etc/elasticsearch/certs/http.p12 -out
/etc/kibana/server.crt -clcerts -nokeys
openssl pkcs12 -in /etc/elasticsearch/certs/http.p12 -out
/etc/kibana/server.key -nocerts -nodes
chown root:kibana /etc/kibana/server.*
chmod g+r /etc/kibana/server.*
ls -la /etc/kibana/
```

8.3 - Activer le SSL dans Kibana

```
nano /etc/kibana/kibana.yml
# Ajouter/modifier vers la ligne 42 :
server.publicBaseUrl: "https://192.168.193.28:5601/"
server.ssl.enabled: true
server.ssl.certificate: /etc/kibana/server.crt
server.ssl.key: /etc/kibana/server.key
```

8.4 - Redémarrer Kibana

```
systemctl restart kibana.service
systemctl status kibana.service
```

Partie IX - Certificat Compatible Navigateurs (Chrome/Edge)

9.1 - Générer le nouveau certificat

```
cd /etc/kibana
openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout server-new.key -out server-new.crt -subj "/CN=192.168.193.28" -addext "subjectAltName=IP:192.168.193.28" -addext "extendedKeyUsage=serverAuth"
```

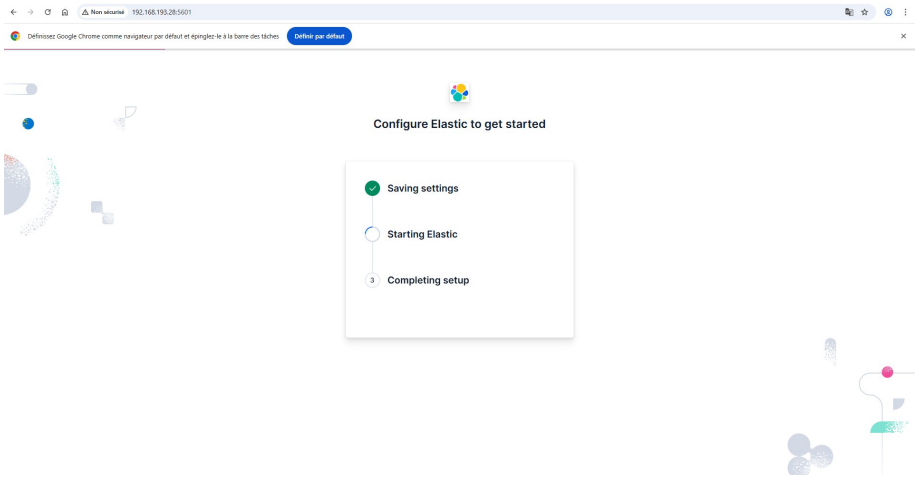
9.2 - Remplacer les anciens certificats

```
mv server.crt server.crt.old
mv server.key server.key.old
mv server-new.crt server.crt
mv server-new.key server.key
```

9.3 - Appliquer les permissions et redémarrer

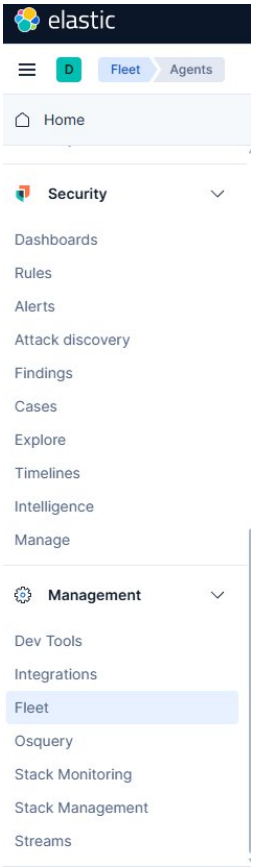
```
chown root:kibana server.*
chmod 640 server.*
systemctl restart kibana
```

Partie X - Résultat Final

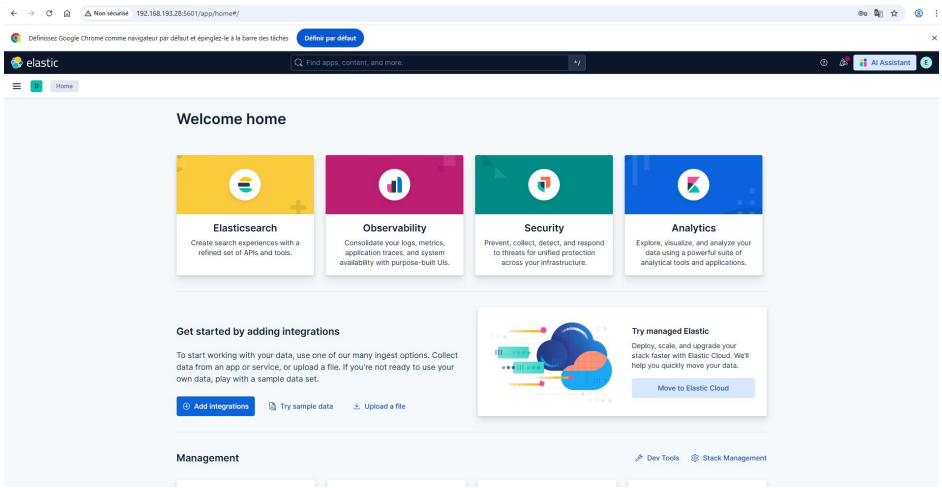


1

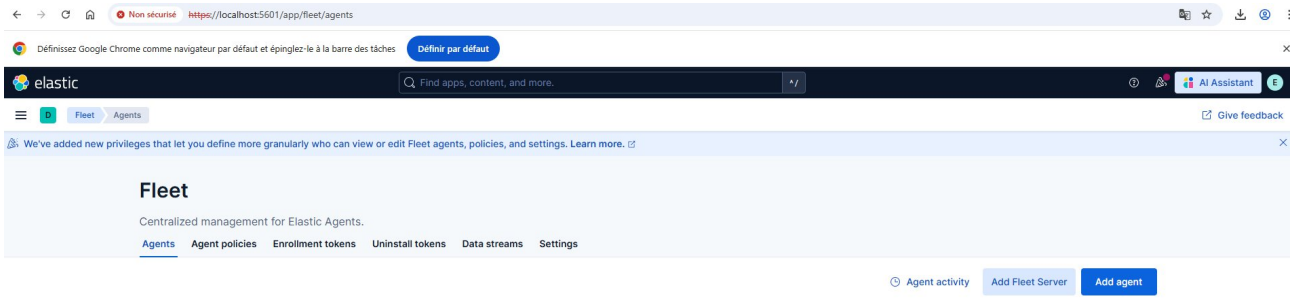
3



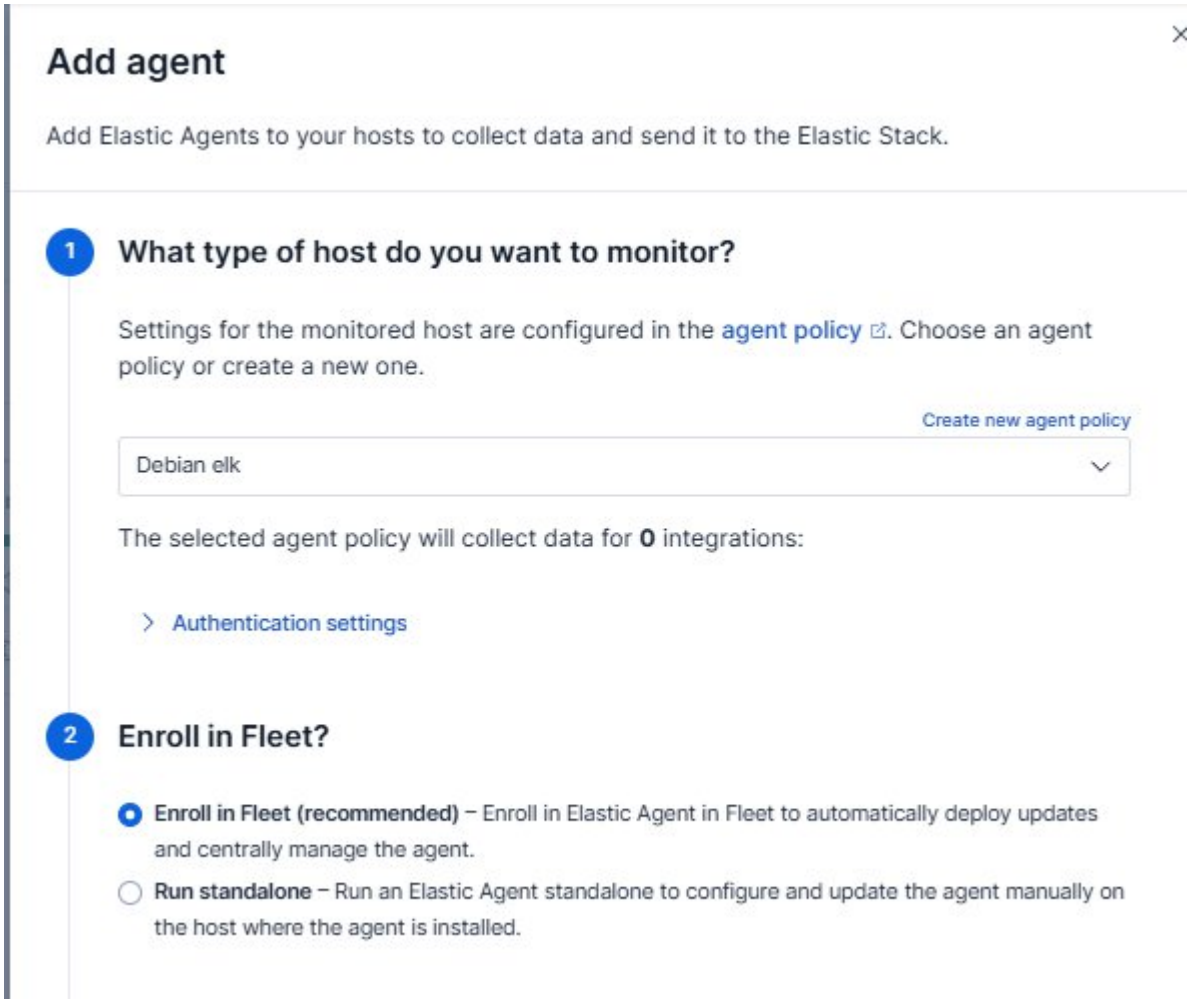
2


Non sécurisé 192.168.193.28:5601/app/home#/
Définissez Google Chrome comme navigateur par défaut et épinglez-le à la barre des tâches
Définir par défaut
elastic
Find apps, content, and more.
AI Assistant
Home
Welcome home
Elasticsearch
Create search experiences with a refined set of APIs and tools.
Observability
Consolidate your logs, metrics, application traces, and system availability with purpose-built UIs.
Security
Prevent, collect, detect, and respond to threats for unified protection across your infrastructure.
Analytics
Explore, visualize, and analyze your data using a powerful suite of analytical tools and applications.
Get started by adding integrations
To start working with your data, use one of our many ingest options. Collect data from an app or service, or upload a file. If you're not ready to use your own data, play with a sample data set.
Add integrations
Try sample data
Upload a file
Try managed Elastic
Deploy, scale, and upgrade your stack faster with Elastic Cloud. We'll help you quickly move your data.
Move to Elastic Cloud
Management
Dev Tools
Stack Management

4


Non sécurisé https://localhost:5601/app/fleet/agents
Définissez Google Chrome comme navigateur par défaut et épinglez-le à la barre des tâches
Définir par défaut
elastic
Find apps, content, and more.
AI Assistant
Fleet
Agents
Give feedback
We've added new privileges that let you define more granularly who can view or edit Fleet agents, policies, and settings. Learn more.
Fleet
Centralized management for Elastic Agents.
Agents
Agent policies
Enrollment tokens
Uninstall tokens
Data streams
Settings
Agent activity
Add Fleet Server
Add agent

5


Add agent
Add Elastic Agents to your hosts to collect data and send it to the Elastic Stack.
1
What type of host do you want to monitor?
Settings for the monitored host are configured in the agent policy. Choose an agent policy or create a new one.
Create new agent policy
Debian elk
The selected agent policy will collect data for 0 integrations:
Authentication settings
2
Enroll in Fleet?
Enroll in Fleet (recommended) – Enroll in Elastic Agent in Fleet to automatically deploy updates and centrally manage the agent.
Run standalone – Run an Elastic Agent standalone to configure and update the agent manually on the host where the agent is installed.

6

Quick Start | Advanced

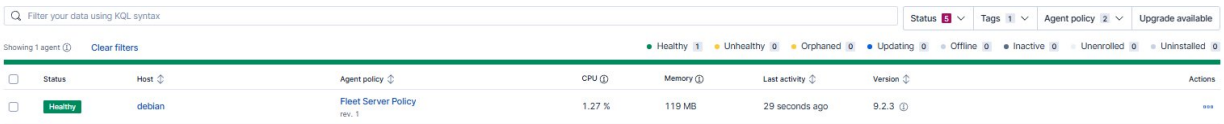
Install Fleet Server agent on a centralized host so that other hosts you wish to monitor can connect to it. In production, we recommend using one or more dedicated hosts. For additional guidance, see our installation docs.

Linux aarch64 | MacOS aarch64 | DEB aarch64 | RPM aarch64 | ... 1

Windows x86_64
Windows MSI
Linux x86_64
MacOS x86_64
DEB x86_64
RPM x86_64

⚠ We recommend using the installers (TAR/ZIP) over s PM/DEB) because they provide the ability to upgrade your agent

```
curl -L -O https://artifacts.elastic.co/downloads/ t/elast
sudo ELASTIC_AGENT_FLAVOR=servers dpkg -i elastic- deb
sudo systemctl enable elastic-agent
sudo systemctl start elastic-agent
sudo elastic-agent enroll \
  --fleet-server-es=https://192.168.193.28:9200 \
  --fleet-server-service-token=AAEAAWVsYXN0aWMvZmxlZXQtc2VydmVyL3Rva2VuLTE3
  --fleet-server-policy=fleet-server-policy \
  --fleet-server-es-ca-trusted-fingerprint=7fba03f7d9328cd04d762b9f96b86cac
  --fleet-server-port=8220
```



✓ **Résultat :** *La suite Elastic est installée. Accès à Kibana via https://192.168.193.28:5601 avec le token d'enrôlement ou les identifiants configurés.*

— Fin de la documentation —